

Template Week 2 – Logic

Student number: 583809

Assignment 2.1: Parking lot

Which gates do you need?

Het bord moet alleen **full** aangeven als alle 3 plaatsen bezet zijn, dus dat betekend

$P1 = 1$

$P2 = 1$

$P3 = 1$

$F = 1$

$F = P1 \wedge P2 \wedge P3$.

Je moet dus een AND poort kiezen want die geeft true (1) terug als alle drie de parkeerplaatsen bezet zijn.

Complete this table

Parking lot 1	Parking lot 2	Parking lot 3	Result (full)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Assignment 2.2: Android or iPhone

Which gates do you need?

De verkoper mag maar een iPhone kiezen, dus een of het ander daardoor gebruik je dan een XOR gate

Complete this table

Android phone	iPhone	Result (Phone in possession)
0	0	0
1	0	1

0	1	1
1	1	0

Assignment 2.3: Four NAND gates

Complete this table

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

How can the design be simplified?

Deze nand poorten kan vervangen worden door een XOR poort daardoor word het design simpel en kleiner

Assignment 2.4: Getting to know Logisim evolution

Screenshot of the design with your name and student number in it:

Bestand Bewerken Project Simuleren FPGA Venster Help main van Zonder titel · Logisim-evolution v4.0.0 [NIET OPGESLAGEN]

Ontwerpen Simuleren

- *Zonder titel
 - main
 - Bedrading
 - Poorten
 - Plexers
 - Rekenen
 - Geheugen
 - Input/Output
 - TTL
 - TCL
 - BFH-megafunctions
 - Input/Output Extra
 - System On a Chip

Eigenschappen Toestand

speld (80,120)

FPGA ondersteuning	Ondersteund
Orientatie	→ Oosten
Type	Input
Woordbreedte	1
Behavior	Simple
Label	HDL vereist
Plaats van het label	← Westen
Label Lettertype	SansSerif Vet 16
Talstelsel	Binair
Verschijsning	Pijlvormen
Opstartwaarde:	0x0

fio

Bestand Bewerken Weergeven

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git Display

and

t (210,140)

Ondersteund

→ Oosten

1

Middel

2

0/1

SansSerif Vet 16

Nee

Nee

fio

Bestand Bewerken Weergeven

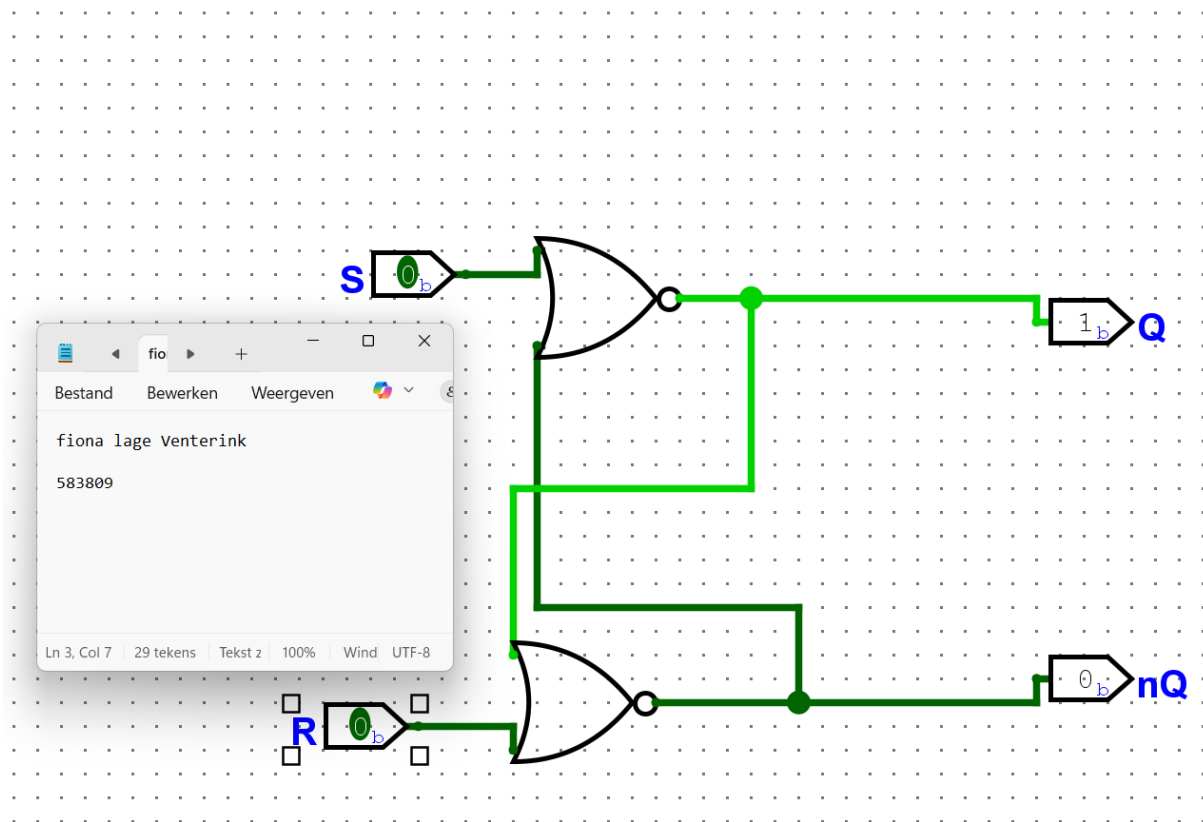
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Assignment 2.5: SR Latch

Screenshot SR Latch in Logisim with your name and student number:



Assignment 2.6: Vending Machine

Screenshot Vending Machine in Logisim with your name and student number:


```
1 // Online Java Compiler
2 // Use this editor to write, compile and run your Java code online
3
4 public class Main { public static void main(String[] args) {
5     int number = 4;
6     if ((number & 1) == 1) System.out.println("oneven");
7     else System.out.println("gelijk"); } }
```

Output

gelijk

=== Code Execution Successful ===

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```
1 // Online Java Compiler
2 // Use this editor to write, compile and run your Java code online
3
4 public class Main { public static void main(String[] args)
5 {
6     int[] numbers = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16};
7     for (int number : numbers) { if (number > 0 && (number & (number - 1)) == 0)
8         System.out.println(number + " is a power of 2"); else System.out.println
9         (number + " is NOT a power of 2"); }
10 }
11 }
```

Output

0 is NOT a power of 2
1 is a power of 2
2 is a power of 2
3 is NOT a power of 2
4 is a power of 2
5 is NOT a power of 2
6 is NOT a power of 2
7 is NOT a power of 2
8 is a power of 2
9 is NOT a power of 2
10 is NOT a power of 2
11 is NOT a power of 2
12 is NOT a power of 2
13 is NOT a power of 2
14 is NOT a power of 2
15 is NOT a power of 2
16 is a power of 2

=== Code Execution Successful ===

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Ln 3, Col 7 29 tekens Tekst z 100% Wind UTF-8

```
1 // Online Java Compiler
2 // Use this editor to write, compile and run your Java code online
3
4 public class Main { public static void main(String[] args)
5 {
6     final int READ = 4;
7     final int WRITE = 2;
8     final int EXECUTE = 1;
9     int userPermissions = 7; // rwx
10    if ((userPermissions & READ) != 0) System.out.println("User has read permissions");
11    else System.out.println("User can't read. No permissions."); } }
```

Output

User has read permissions

=== Code Execution Successful ===

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Ln 3, Col 7 29 tekens Tekst z 100% Wind UTF-8

```
1 // Online Java Compiler
2 // Use this editor to write, compile and run your Java code online
3 public class Main { public static void main(String[] args)
4 {
5     final int READ = 4;
6     final int WRITE = 2;
7     final int EXECUTE = 1;
8     int userPermissions = 0;
9     userPermissions = userPermissions | READ | EXECUTE;
10    System.out.println("User permissions: " + userPermissions); } }
```

Output

User permissions: 5

=== Code Execution Successful ===

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Ln 3, Col 7 29 tekens Tekst z 100% Wind UTF-8

```

1 // Online Java Compiler
2 // Use this editor to write, compile and run your Java code online
3 public class Main { public static void main(String[] args)
4 {   final int READ = 4;
5     final int WRITE = 2;
6     final int EXECUTE = 1;
7     int userPermissions = 6;
8     userPermissions = userPermissions ^ WRITE;
9     System.out.println("User permissions: " + userPermissions); } }

```

Output

User permissions: 4

=== Code Execution Successful ===

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```

1 // Online Java Compiler
2 // Use this editor to write, compile and run your Java code online
3 public class Main { public static void main(String[] args)
4 {   int number = 5;
5     number = ~number + 1;
6     System.out.println("Number: " + number);
7 }
8 }
9 }

```

Output

Number: -5

=== Code Execution Successful ===

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```

1 // Online Java Compiler
2 // Use this editor to write, compile and run your Java code online
3 public class Main { public static void main(String[] args)
4 {   int number = 10;
5     System.out.println("Decimal integer: " + number);
6     String binary = Integer.toBinaryString(number);
7     String octal = Integer.toOctalString(number);
8     String hexadecimal = Integer.toHexString(number);
9     System.out.println("Binary representation: " + binary);
10    System.out.println("Octal representation: " + octal);
11    System.out.println("Hexadecimal representation: " + hexadecimal); } }

```

Output

Decimal integer: 10

Binary representation: 1010

Octal representation: 12

Hexadecimal representation: a

=== Code Execution Successful ===

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Assignment 2.8: Java Application Bit Calculations

Create a java program that accepts user input and presents a menu with options.

1. Is number odd?
2. Is number a power of 2?
3. Two's complement of number?

Implement the methods by using the bitwise operators you have just learned.

Organize your source code in a readable manner with the use of control flow and methods.

Keep this application because you need to expand it in week 6 for calculating network segments.

Paste source code here, with a screenshot of a working application.

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // 1. Vraag welk getal de gebruiker wil controleren
        System.out.print("Voer een getal in: ");
        int number = scanner.nextInt();

        // 2. Toon het menu
        System.out.println("\n---Menu---");
        System.out.println("1. Is het getal oneven?");
        System.out.println("2. Is het getal een macht van 2?");
        System.out.println("3. Two's complement van het getal?");
        System.out.print("Maak een keuze: ");

        int choice = scanner.nextInt();

        // 3. Voer de juiste check uit
        switch (choice) {

            case 1:
                // Oneven check met bitwise AND
                if ((number & 1) == 1)
                    System.out.println(number + " is oneven");
                else
                    System.out.println(number + " is even");
                break;

            case 2:
                // Macht van 2 check met bitwise AND
                if (number > 0 && (number & (number - 1)) == 0)
                    System.out.println(number + " is een macht van 2");
                else
                    System.out.println(number + " is GEEN macht van 2");
                break;

            case 3:
                // Two's complement met bitwise NOT
                int twosComp = ~number + 1;
                System.out.println("Two's complement van " + number + " = " + twosComp);
                break;
        }
    }
}
```

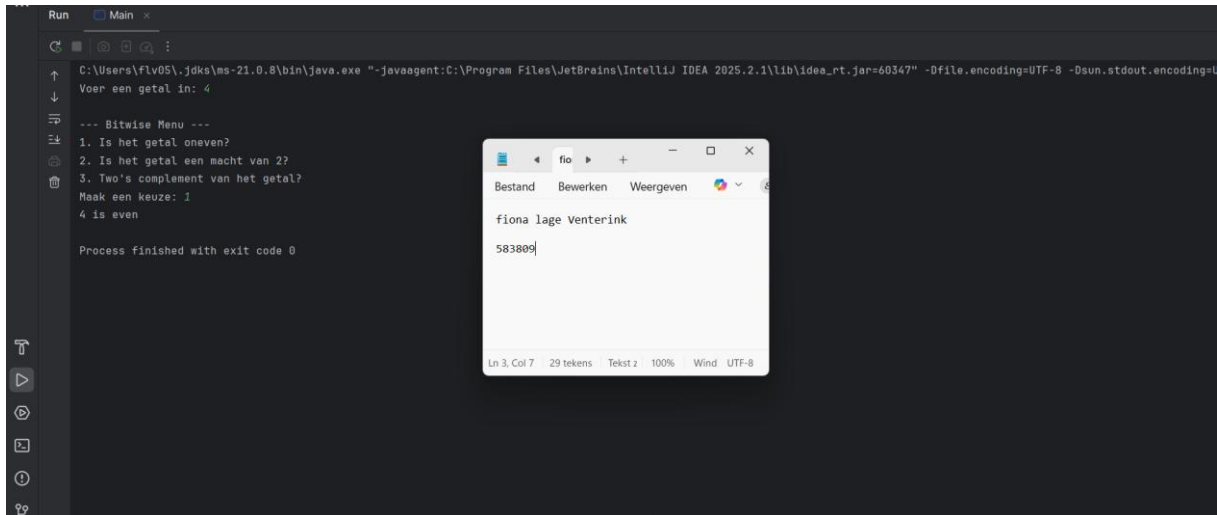


```

        default:
            System.out.println("Ongeldige keuze");
        }

    scanner.close();
}
}

```



Ready? Then save this file and export it as a pdf file with the name: [week2.pdf](#)